Basic Fish Anatomy

By

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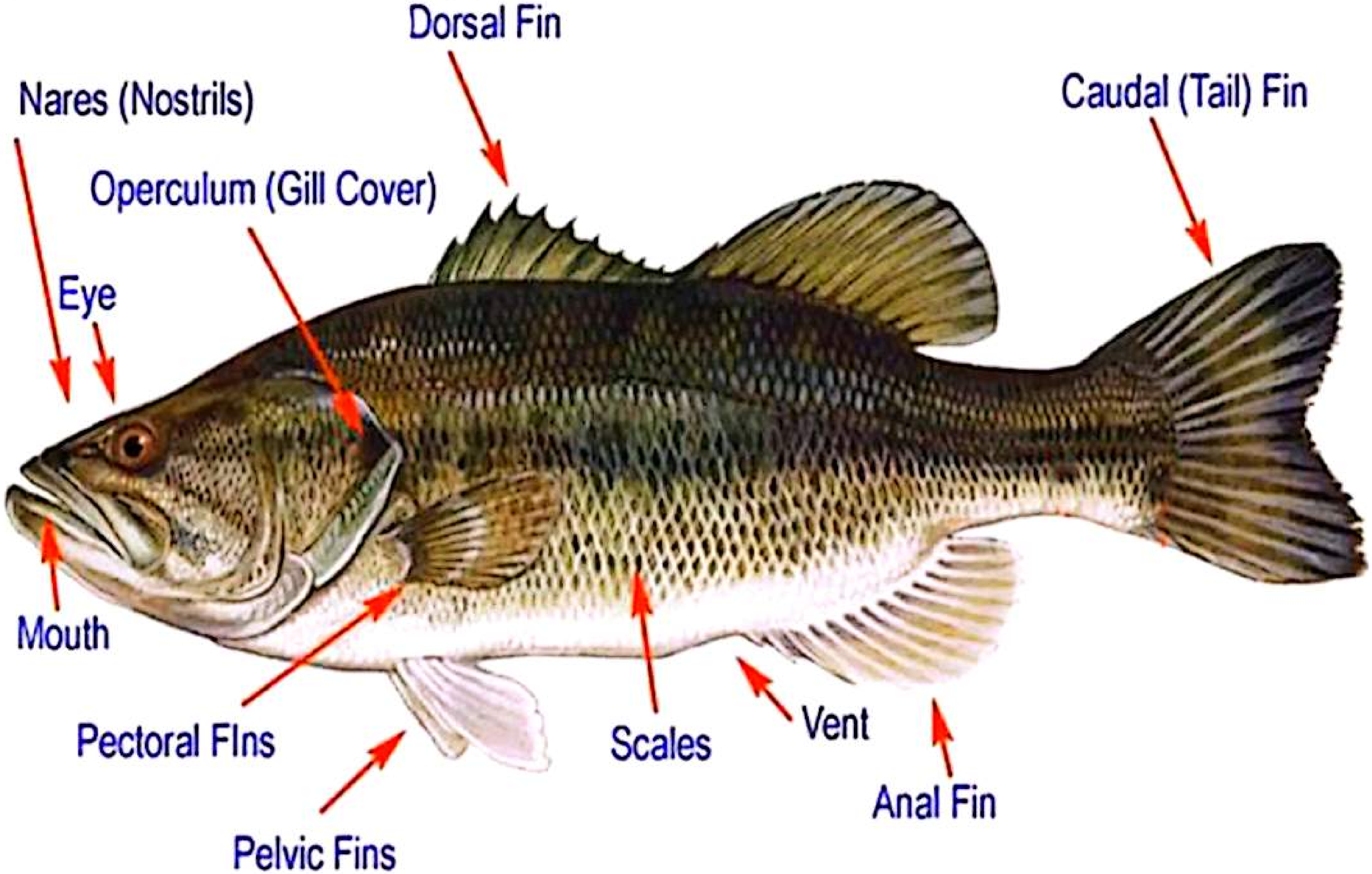


EXTERNAL ANATOMY

Digestive System of fish

- **Adaptation & specialization of digestive tract** to suit specific diets are often very distinctive.
- **These adaptions include:**

❑ **Length of digestive tube is a major difference**

➢ **Digestive tract of herbivorous fish:** very longer than of carnivores.

❑ **Specialization related to diet includes also** the mouth, tooth the presence and the number of diverticulate (caeci).

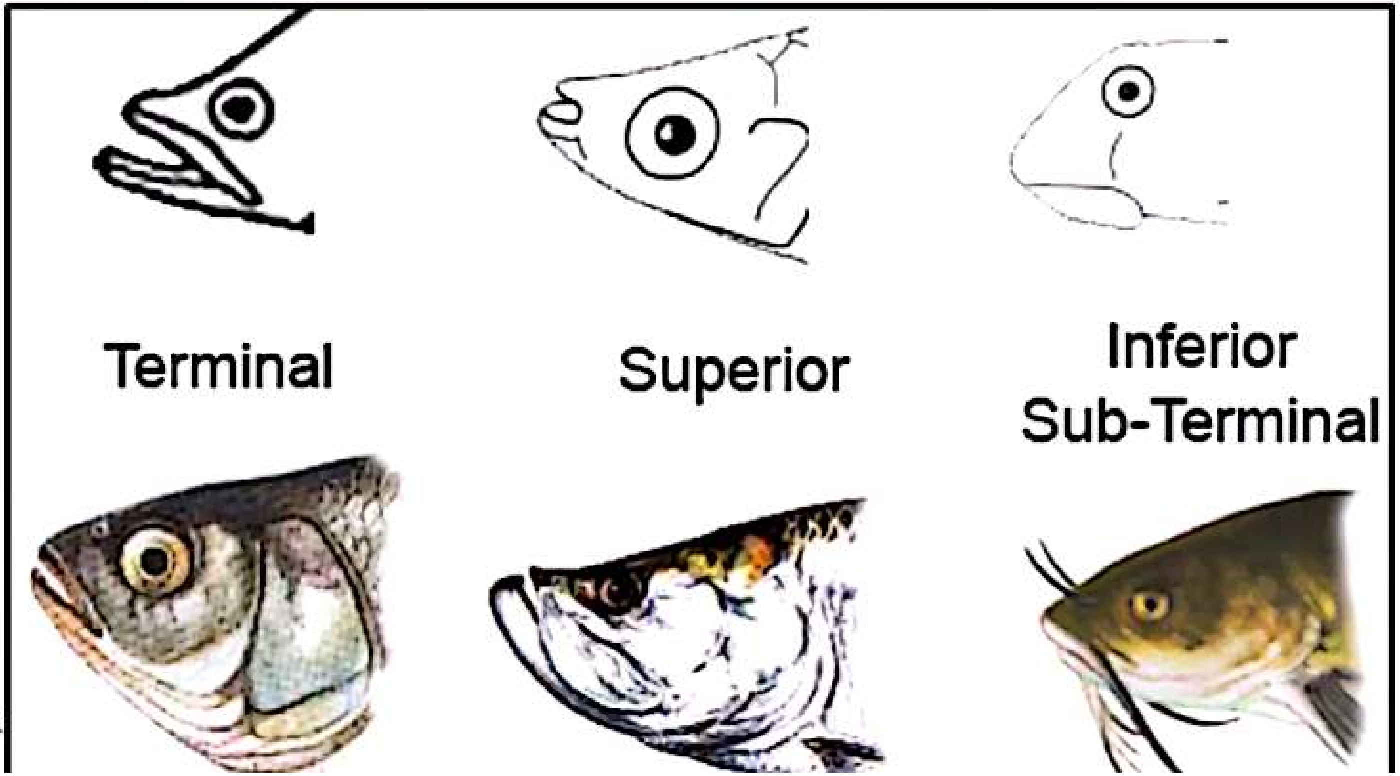
1- Mouth

- Forms beginning of digestive tract;
- **Mouth & buccal cavity** serve digestive &respiratory functions.
- **Digestive functions** confined to selection, seizure & orientation of food toward esophagus and then to stomach.
- **Chewing & pre-digestive process** which occurred in mammals is not usually of mouth function of teleost except in few higher herbivorous fish.
- **Position of fish moth explains the feeding habits:**

Types of fish mouth

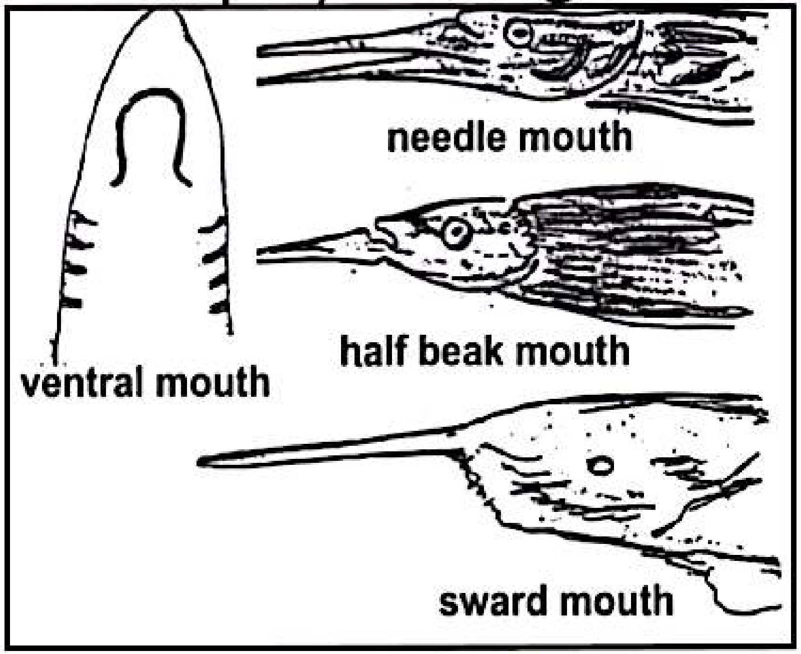
A) According to mouth position and food habit:

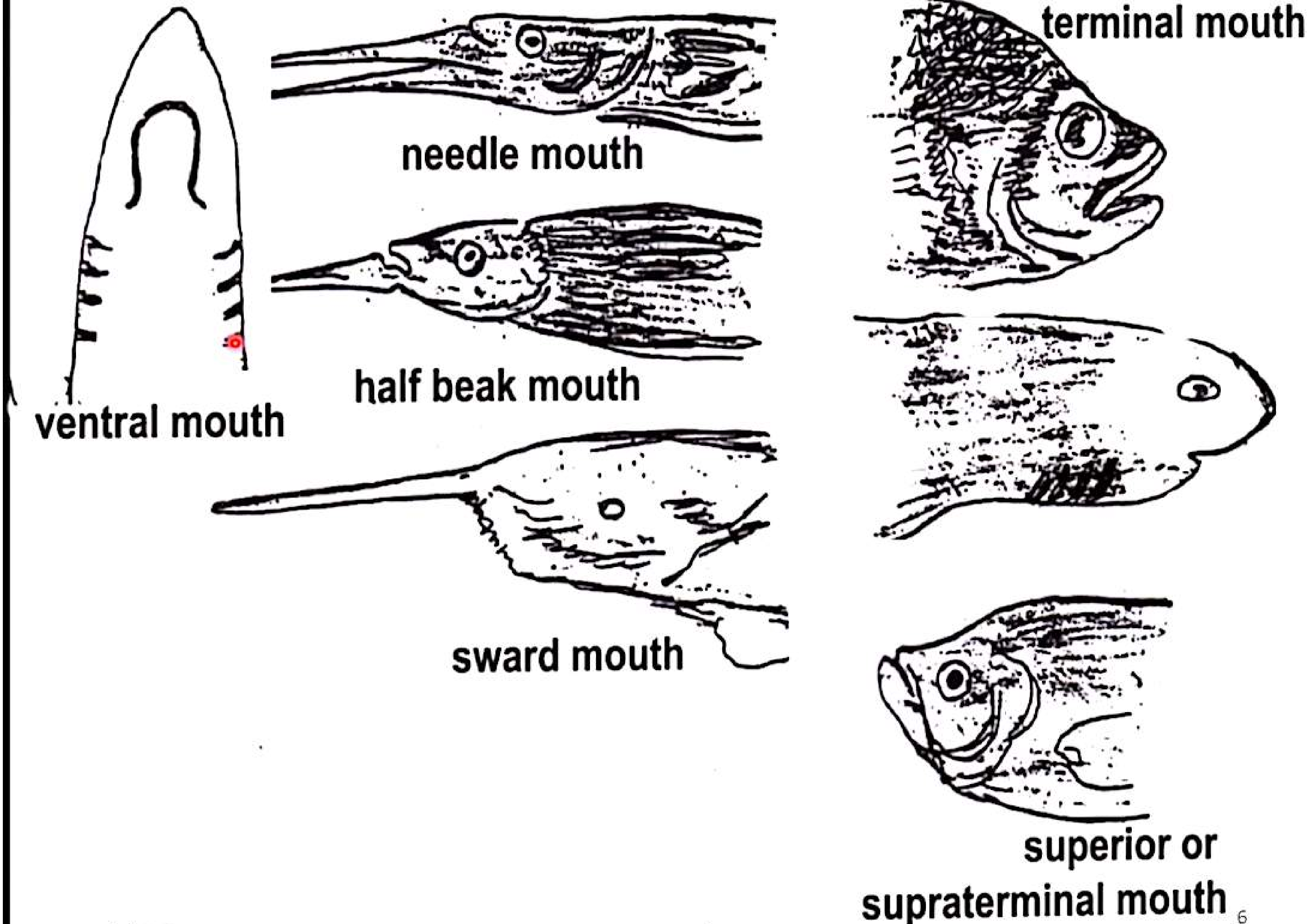
Mouth position	Surface feeder	Bottom feeder
Superior	√	X
Vertical	√	X
Inferior	X	√
Terminal	√	√
Subterminal	X	√



B) According to jaw shape:

1- Opposing jaws	Sharks & bony fishes (Tilapia, Carp &Catfish)	
2- Non opposing jaws	**a. Half beak mouth** • Lower jaw: Long and spear-like • Upper jaw: very short	**b. Sword mouth** • Upper jaw: very long and sword like • Lower jaw: markedly short
	Both jaws are greatly elongated & have sharp teeth As: agnnatha lamprey and hagfishes.	




terminal mouth
needle mouth
half beak mouth
ventral mouth
sward mouth
superior or
supraterminal mouth

2- Teeth

- Vary greatly in **location, number & morphology** acc. to **feeding habits and type of fishes.**

❑ ***For example:***

✓ **Carnivorous fishes**: have long sharp canine teeth suitable for holding captured prey.

✓ **Herbivorous fishes**: have filiform and /or incisor teeth adapted for cutting algae as **Damsel Fish**

✓ **Bony fishes:** have mandibular, maxillary, palatine, lingual and pharyngeal teeth.

✓ **Cartilaginous fishes**: have only maxillary and mandibular teeth.

Some fish not have teeth as Rabbit Fish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3- Tongue

- Generally, the fish tongue is **absent**
- If present, **very small elevation** from mouth floor supported by CT & cartilage and have no muscular tissue (i.e can't move alone)
- Fish tongue is used **only for taste.**

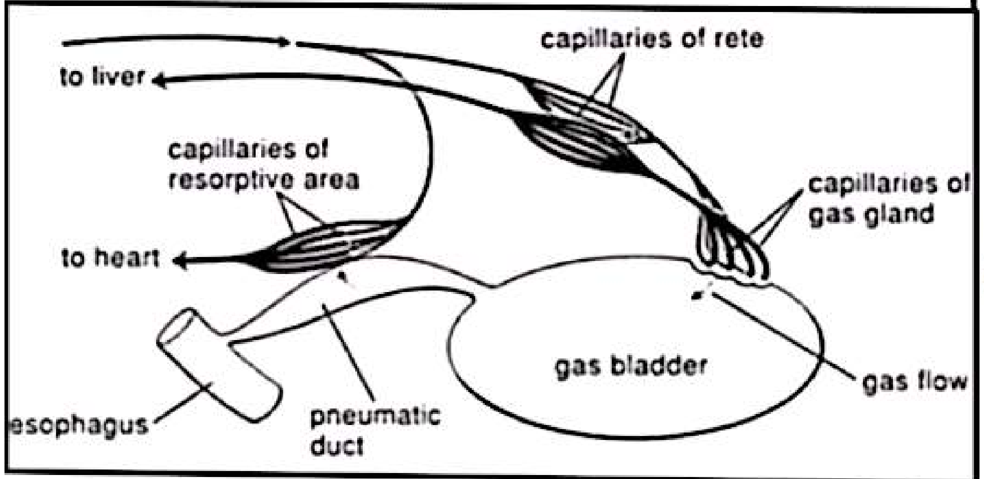
4- Pharynx

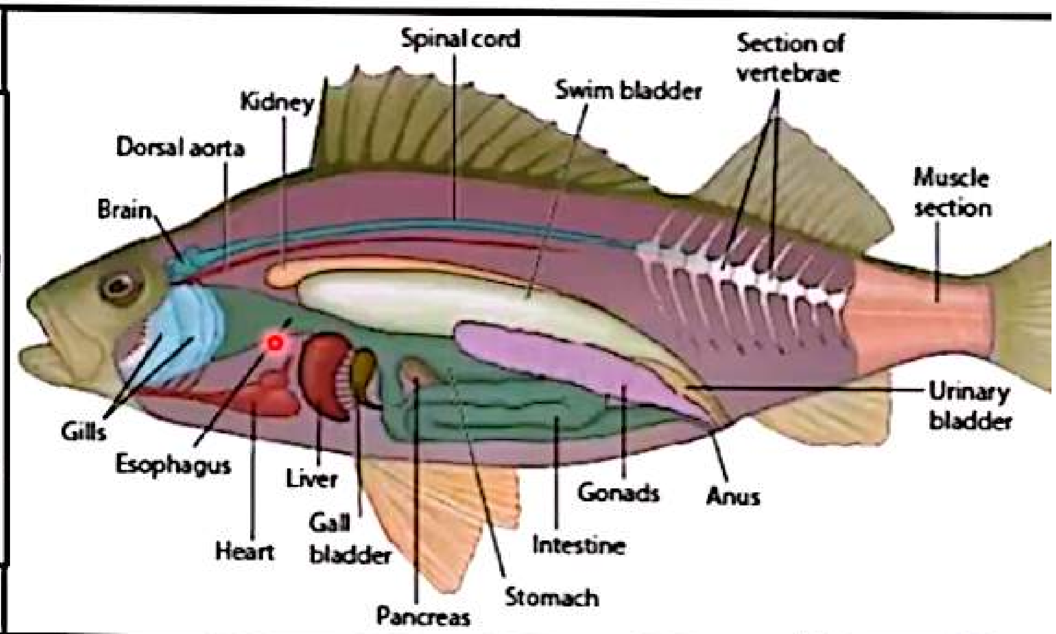
- Follows oral cavity and the two form of oropharynx (buccal cavity).
- It's perforated by the **branchial clefts** between the gill arches.

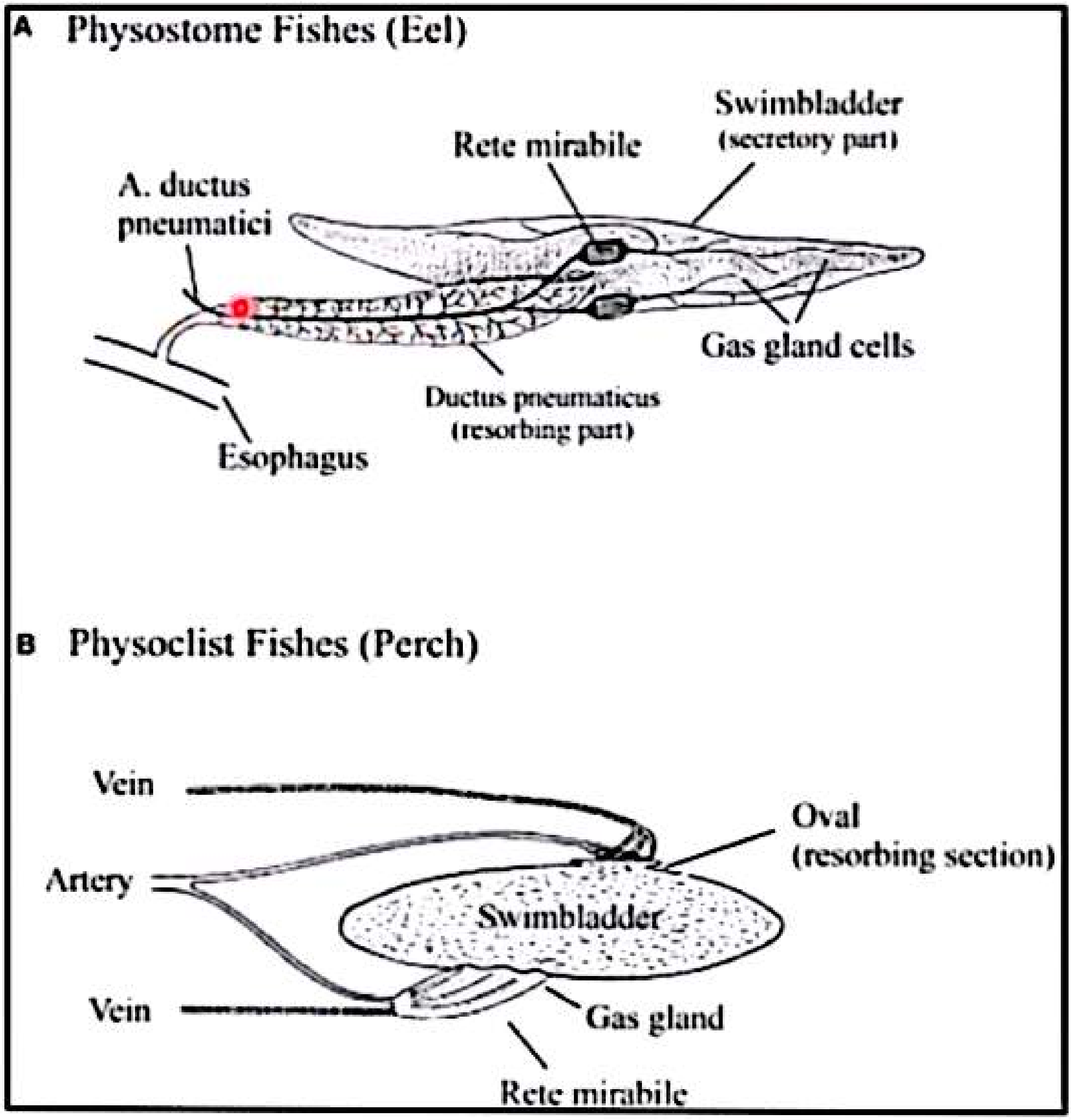
N.B. Fungiform and filiform papillae may be found in the oral cavity and pharynx

5- esophagus

- ✓ A short and straight tube connects pharynx to stomach.
- ✓ Separated from stomach by slight constriction, but two may be distinguished histologically by
  - ➢ Squamous epithelium of esophagus replaced by columnar epithelium of stomach.
  - ➢ Presence of gastric glands in the stomach.
- ✓ It connects with air bladder through ***pneumatic duct.***
- ✓ Occasionally have taste buds.

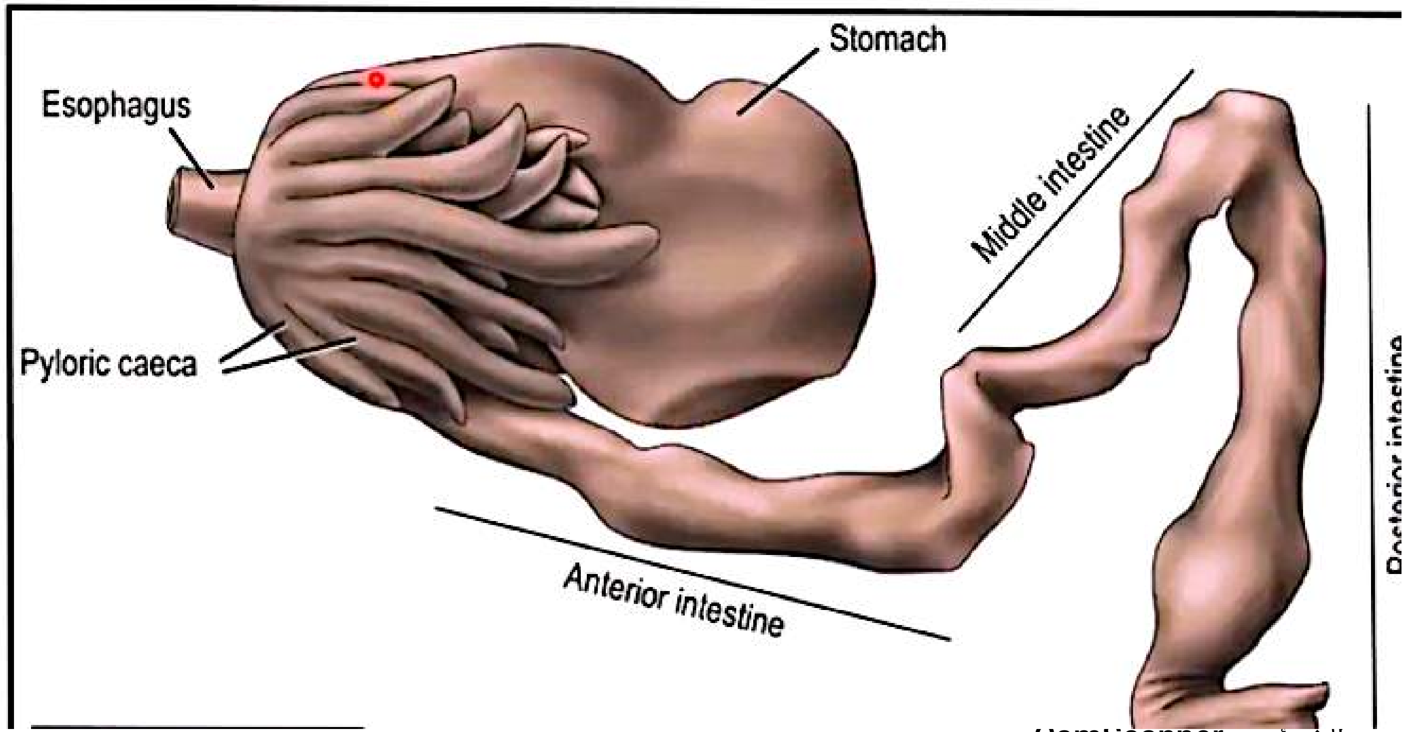





A Physostome Fishes (Eel)
Swimbladder
(secretory part)
Rete mirabile
A. ductus
pneumatici
Gas gland cells
Ductus pneumaticus
(resorbing part)
Esophagus
B Physoclist Fishes (Perch)
Vein
Oval
(resorbing section)
Artery
Swimbladder
Vein
Gas gland
Rete mirabile

6- Stomach

- ✓ **U- or V-shaped**, highly distensible musculo-membrane sac with numerous mucosal folds.
- ✓ **Cardia** demarcates change from esophageal striated muscles to stomach smooth muscle.
- ✓ **Gastric mucosa** is very mucoid & rich in numerous glands at base of folds.
- ✓ **Mugile** has heavy muscular stomach which called ***Gizzard stomach*.**
- ✓ **Carp** has **no stomach**; so food pass direct from esophagus to intestine

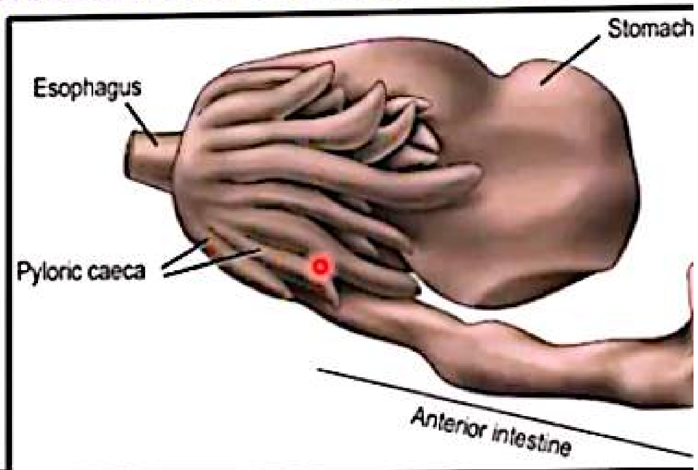


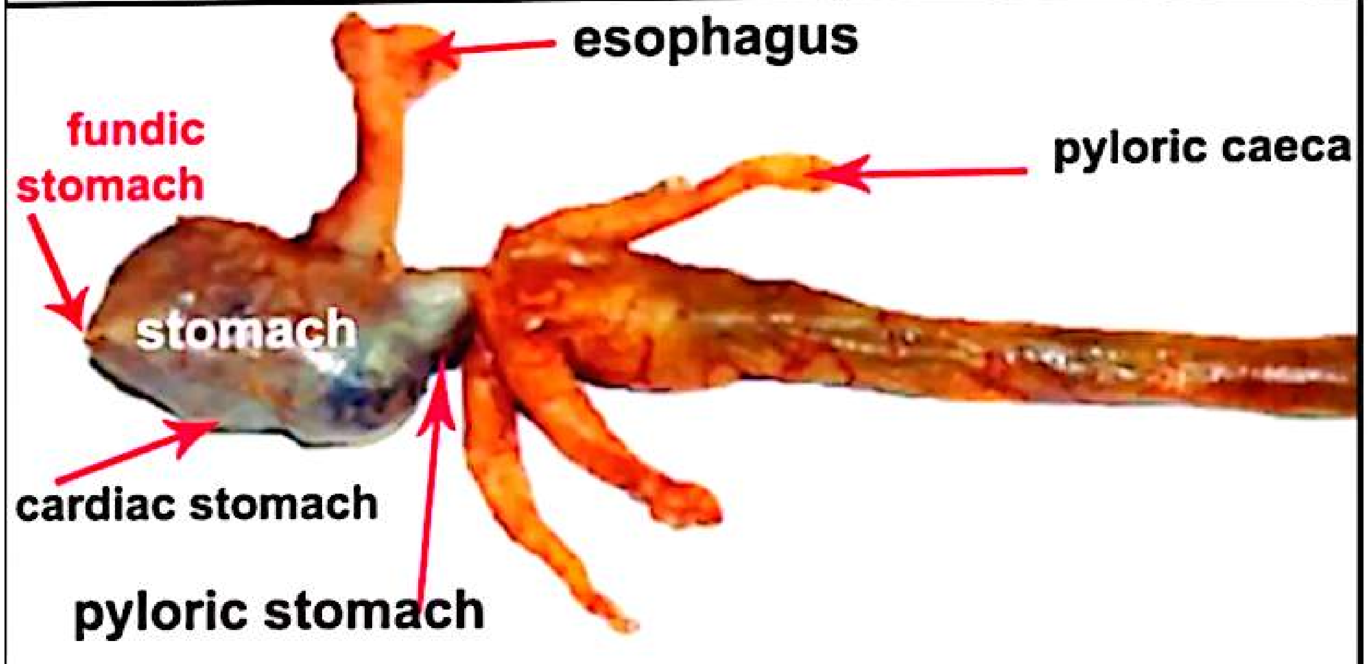
Pyloric caeca

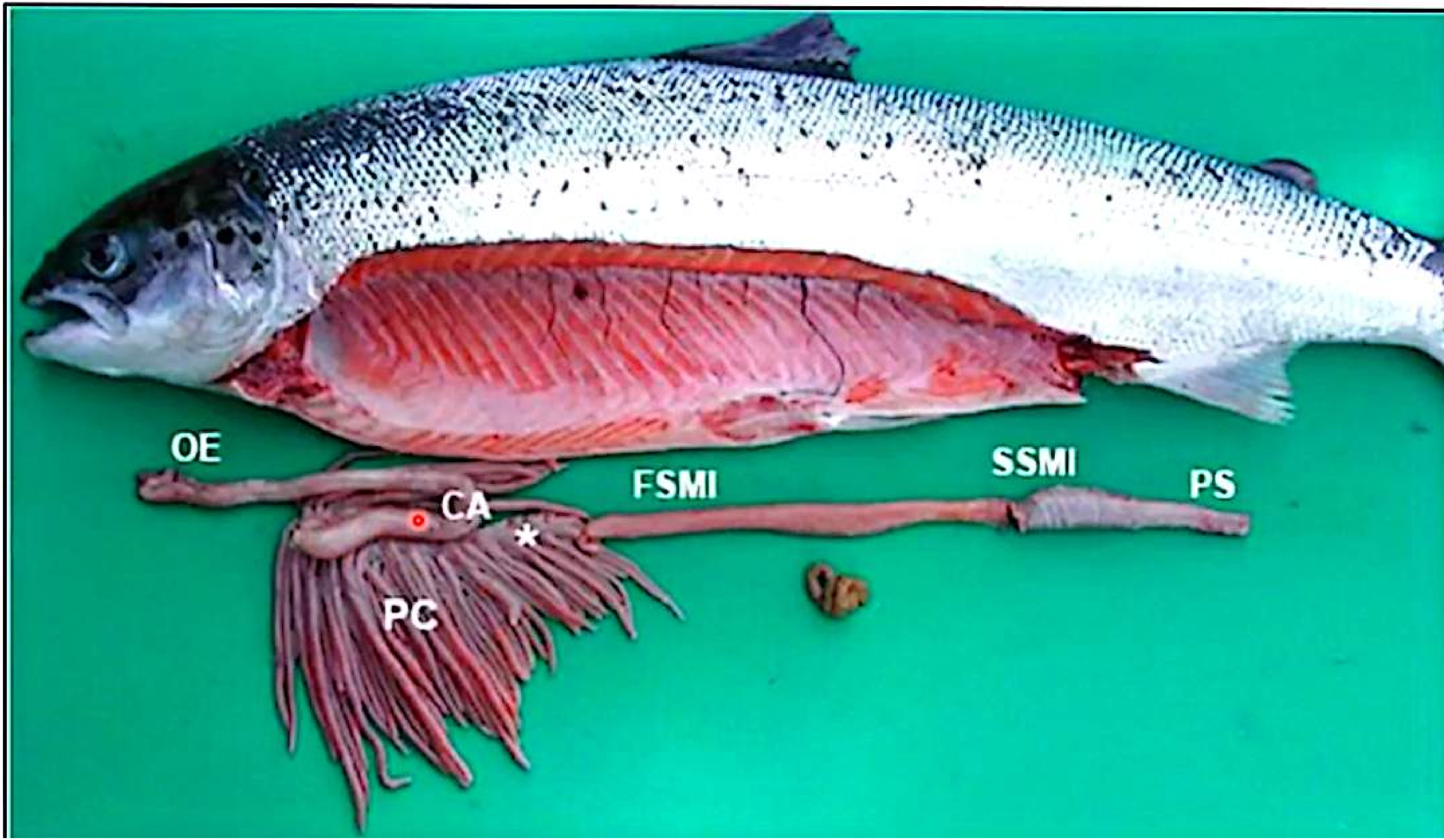
- Blind ending pouches (70 or more) arise from distal part of stomach (pyloric region) and pyloric valves between stomach &duodenum.
- Their histological features resemble those of intestine rather than the stomach.
- No pyloric caeca in Catfish.

Function:

- Secretes enzymes help in digestion
- Has absorption function



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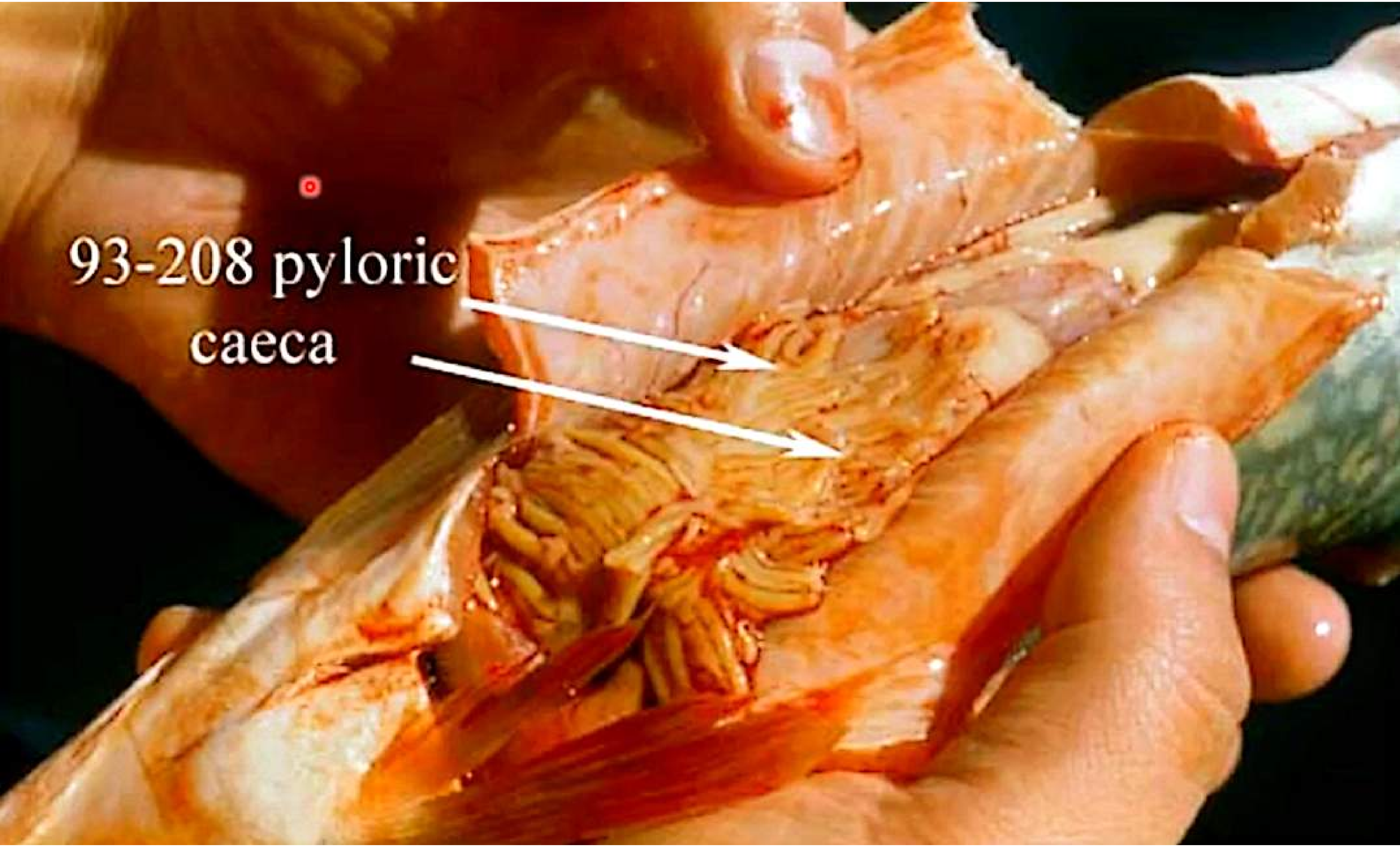

93-208 pyloric
caeca

7- Intestine

- ✓ Intestine of most fish is simple musculo-membranous tube which does not increase in diameter to form colon posteriorly.
- ✓ **Relative length of intestine** depend mainly up on nature of diet, as:
  - o Predaceous bony fish: living chiefly on animal diet, intestine is short (1-3 loops)
  - o Veritable feeders fish: intestine is long with many loops therefore shape of intestinal mass may be straight, sigmoidal or coiled.
- ✓ **Ducts or combined ducts** from pancreas & liver enter intestine just caudal to pyloric valve.

7- Intestine

A- Small intestine (duodenum, jejunum, and ileum):

Length &shape depend on the nature of the diet:

Fish	Length	Shape of intestine
Carnivorous	Short with one to three loops	Straight
Herbivorous	Long with many loops	Coiled
omnivorous	Intermediate length	Sigmoid

B- Large intestine (rectum)

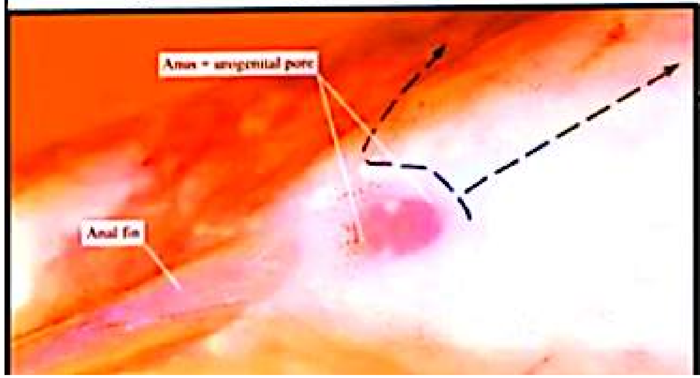
✓ **Rectum is distinguished from the small intestine by:**

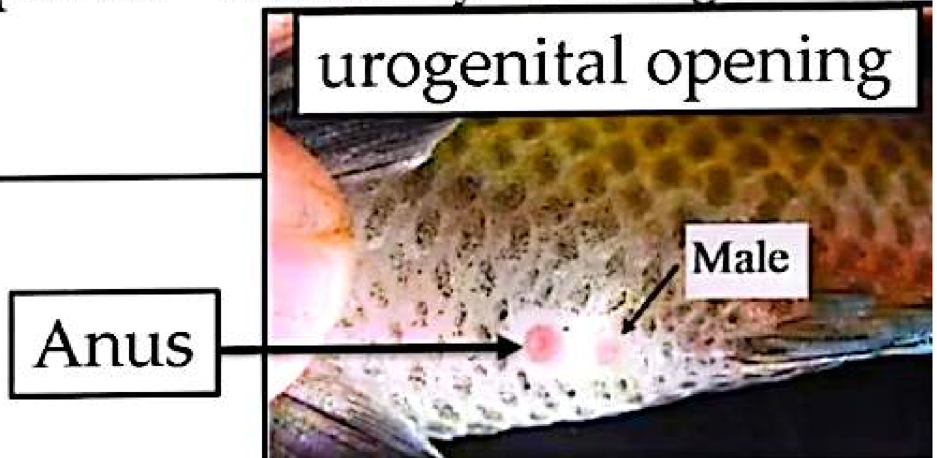
1. Its straight course
2. Its wider diameter

✓ Presence of **cecal diverticulum** (in some fishes) and **rectal gland**

C- Anus

- Exterior opening of the digestive tract of bony fish
- It is located on the ventral (belly) side of the fish,
- Typically, it is located just cranial to the anal fin and posterior to the pelvic fins
- **Bony fish** have a separate anus (for the digestive system) and a urogenital opening (for the urinary and reproductive systems)
- **in cartilaginous fish**, no anus and so rectum ends caudally in cloaca with urogenital opening, that opened externally through vent opening.





8- Liver

- Reddish brown in carnivorous and lighter brown in herbivorous.
- <u>Lobation</u>:
  1. *One lobe (unilobed)*: as in salmon (trout)
  2. *Two lobes (bilobed)*: in most bony fish, such as tilapia nilotica.
     a) *Left lobe* forms with the pancreas, a compound organ named *hepatopancreas*
     b) *Right lobe* connects with the gall bladder.
  3. *Three lobes (trilobd)*: as in Mackerel

NB:

✓ **<u>liver of some fish, such as Catfish, T. nilotica, Mugile</u>** has a gall bladder that contains greenish yellow bile.

✓ **<u>liver is an organ notable for its sensitivity to a great variety of environmental factors</u>**, so it is a biomarker of environmental pollution.

8- Liver

In case of hepatopancreatic fish

(Catfish, T. nilotica, Mugil, Spiny dog fish, Sea scorpion, Sea-bass, Hake, Sea eel)

↓

Fish have evolved some organs that mammals have not

↓

One of them is the ***exocrine pancreatic tissue in the liver***

↓

This exocrine pancreatic tissue is formed around the portal vein

During ontogenesis, it penetrates deeply into the liver parenchyma,

Its proportion depends on the species of fish.

- **Exocrine pancreatic tissue** is mainly separated from hepatic parenchyma by a thin layer of C.T.

Duct system

- **Biliary system** occurs by anastomosis of intercellular bile canaliculi (Intrahepatic) to form a typical hepatic (bile) duct.
- Right hepatic (bile) duct: come from right lobe
- Left hepatic (bile) duct: come from left lobe
- Cystic duct: come from gall bladder
- Common bile duct (open in duodenum)= 1+2+3

9- Pancreas

✓ **Unlike in mammals:**

➢ In some fish, the pancreas is **diffuse or "disseminated"**

❖ It is scattered in many other organs and places, including the mesentery, pyloric caeca, subcapsule of spleen, and substances of the liver.

❖ Particularly in teleosts (ray-finned fish)

➢ In some fish: is compact, but can be composed of lobules, in which case it mostly consists of two lobules.

✓ The pancreatic duct: either

- Joins the liver by the common bile duct or
- Opens separately in the duodenum.

✓ Histologically, the pancreas is formed from:

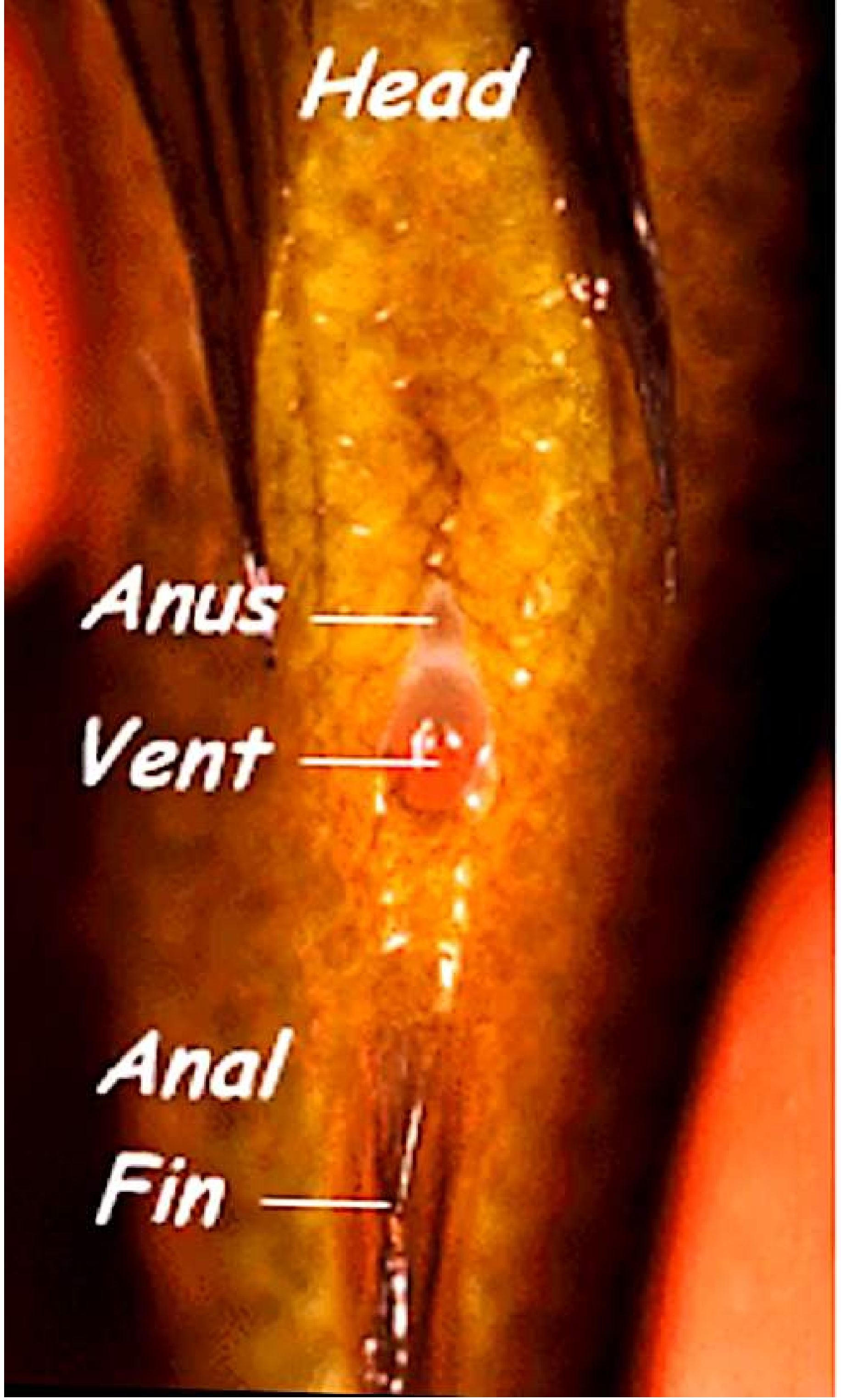
- **Exocrine part**: secretes digestive juices
- **Endocrine part**: secretes the hormones insulin and glucagon

9- Peritoneum

- It has two layers:

1. **Visceral layer**: covers abdominal organs
2. **Parietal layer;** lines the peritoneal cavity

- The two layers are black, especially in herbivorous bony fishes


Head
Anus
Vent
Anal
Fin